

M GRISHMA

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Summary

ML Infrastructure Engineer specializing in LLM inference optimization and GPU-accelerated deep learning systems. Built a production-grade autograd engine in C++/CUDA from scratch — covering VJP-based reverse-mode AD, dynamic computation graphs, lazy gradient materialization, and PTX-level kernel optimization — achieving **52K tokens/sec** with **99% GPU active time** on consumer hardware. Experienced in custom CUDA kernel development, inference latency reduction, and turning research into production-grade ML systems. Proficient with PyTorch internals, CUDA programming model, and modern AI developer tooling including Claude Code, Cursor, and Antigravity.

Education

B.Tech in Computer Science Engineering | Presidency University, Bangalore Graduation: July 2025
BITS Pilani WILP — Ongoing enrollment

Technical Skills

- **Languages:** C++ (Advanced: C++17/20), CUDA C++, Python, C
- **LLM Inference & Optimization:** Inference latency optimization, throughput optimization, KV cache, PagedAttention (vLLM), FlashAttention, TensorRT, ONNX, quantization (INT8/FP16/BF16), operator fusion, kernel fusion
- **AI & Systems:** Automatic Differentiation (Backpropagation, VJP), Computational Graphs, Lazy Gradient Materialization, SavedVariable Mechanics, Compiler Frontend Design (IR), MLIR/LLVM, Deep Learning Framework Architecture, PyTorch internals, HPC
- **CUDA & GPU Programming:** CUDA Kernels, PTX Inspection & Optimization, CUDA Streams, CUDA Graphs, cuBLAS, warp-level primitives, memory bandwidth optimization, H2D/D2H Transfer Optimization, shared memory optimization
- **CPU Optimization:** AVX2, OpenMP, mixed precision training (FP16/BF16), distributed training concepts, NCCL
- **Profiling & Debugging:** Nsight Systems, Nsight Compute, Intel VTune, AMD uProf, GDB, roofline model analysis
- **Frameworks & Tools:** PyTorch, Git, CMake, Make, Docker, Kubernetes, Flask, MySQL, MongoDB
- **Developer Tooling:** Claude Code, Codex, Cursor, OpenClaw, Antigravity, workflow automation

Experience

AI Systems Engineer | BluBridge, Chennai, India

July 2025 – March 2026

- **Team lead** — architected and built a production C++/CUDA deep learning framework; scoped and delegated four parallel workstreams: ABI adapter, gradient checkpointing, MLIR/LLVM compiler, and autograd hooks infrastructure
- Designed the core autograd engine — VJP-based reverse-mode AD over a dynamic computation graph with AutogradMeta + MetaInterface for deferred, on-demand gradient materialization — achieving **52K tokens/sec** on RTX 3060 + Intel i9-14700
- Engineered serial and parallel backward pass execution across CPU and GPU backends; introduced **SavedVariable** to cut peak memory pressure and **cached BFS traversal** to eliminate repeated graph walks per iteration
- Built the full execution pipeline around a custom async dispatcher, sustaining **99% GPU active time** and **68% average SM occupancy** across training workloads
- Used Nsight Systems & Compute, VTune, and uProf to eliminate redundant H2D/D2H transfers; inspected PTX output to identify and apply PTX-level optimizations (loop unrolling, instruction scheduling); optimized backward CUDA kernels for memory access patterns and occupancy

Machine Learning Intern | D4 Insights, Chennai, India

July 2024 – August 2024

- Built and optimized ML models for predictive analytics pipelines; worked on feature engineering and model evaluation for business intelligence applications

ML and Neural Net Intern | Azure Skynets (Remote)

June 2024 – July 2024

- Fine-tuned neural network models for AI applications; applied transfer learning and performance benchmarking to improve inference accuracy and latency metrics

Projects

Cortex: Production C++ Deep Learning Framework | *C++/CUDA, PyTorch-comparable architecture*

- Architected an eager-execution deep learning framework in C++20 with a dynamic computational graph and robust reverse-mode automatic differentiation engine — comparable in design to PyTorch's autograd internals
- Implemented the core Autodiff engine with VJP (Vector-Jacobian Product) rules for 50+ operations; solved a critical broadcasting bug ensuring gradient correctness across all tensor shapes
- Engineered backend-agnostic architecture with a dynamic plugin system; optimized CPU (AVX2, OpenMP) and GPU (cuBLAS, CUDA) backends for maximum throughput; applied kernel fusion and memory bandwidth optimizations
- Designed a JIT compiler frontend lowering the dynamic graph into a linear Intermediate Representation (IR) for compatibility with CUDA Graphs and backend optimizers; applied operator fusion and mixed precision (FP16/BF16) support
- Achieved 52K tokens/sec sustained throughput with 99% GPU active time and 68% SM occupancy on RTX 3060 — profiled and optimized using Nsight Compute and roofline model analysis

Certifications

- **Data Science, AI & Development with Python** — IBM (October 2024)
- **Python for Data Science** — NPTEL (October 2024)
- **Introduction to Artificial Intelligence** — NPTEL (April 2024)
- **Introduction to Supervised & Unsupervised Machine Learning** — Simplilearn (November 2023)